

Optiver Trading Challenge

Trading Challenge: Options Market Making strategy





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1 Introduction

In this trading challenge you will work on an algorithm to quote/market make a set of stock options. The challenge builds upon the concepts covered in the Dual Listing quoting strategy (passive).

Again, your algorithm will take prices from a liquid instrument (the stock) to a set of less liquid instruments (the options) and hedge using the liquid instrument. A difference is that the prices between the traded instruments no longer relate on a one-to-one basis. Another difference is that several options will be quoted simultaneously.

After finishing it, your algorithm will continuously and profitably provide liquidity in the options through inserting limit orders based on the prices of a stock, ensuring to manage positional risk by keeping total outstanding positions delta-hedged.



2 Trading Challenge

There are typically many options listed on a single underlying, e.g. calls and puts of different strikes and expiries. Using the derivative pricing theory of mathematical finance, it can be shown that these option prices do not vary completely independently, but rather all depend on the stock price through a fixed relationship.

Under a specific, and quite strong, set of assumptions, the relationship is described by the Black-Scholes pricing formulas:

$$C = S \cdot N(d_1) - K \cdot e^{-r\tau} \cdot N(d_2),$$

$$P = K \cdot e^{-r\tau} \cdot N(-d_2) - N(-d_1) \cdot S,$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)\tau}{\sigma\sqrt{\tau}},$$

$$d_2 = d_1 - \sigma\sqrt{\tau},$$

With C and P the call and put price, S the stock price, K the strike price of the option, σ the annualized volatility of the stock process, r the risk-free interest rate, and τ the time to expiry in years. For the purpose of this exercise, these Black-Scholes equations (and underlying assumptions) can be assumed to always hold. Further, the annualized volatility of the stock log-returns is fixed at $\sigma = 3.0$ (300%). Note that this volatility is much higher than ordinarily seen on stocks, and set such to ensure the simulated exchange is always lively. The annual, continuously compounded, interest rate is $r = 0.03$ (3%). The challenge reference algorithm contains an example of how calculate the time to expiry τ .

In this challenge, rather than scouting for trade opportunities in existing orders, the algorithm should provide liquidity through quoting two prices around a theoretical price, through adding a so-called “bid-ask credit”. Other market participants (bots and/or other users) can then trade against those prices. The idea is that if a buyer buys from our ask followed by a seller selling to our bid, or vice versa, this leads to a locked in profit for the strategy. As that does not always happen directly, in the meanwhile the positional risks need to be managed.

The largest positional risk to manage is directional movement of the stock, leading to simultaneous movement in the corresponding options according to their deltas. This “delta risk” of an options portfolio can be hedged through performing offsetting trades in the stock itself.

Implement the following trading strategy based on the provided sample code in the `options_mm_template.py` in the ‘options_market_making’ directory in Optibook.



2.1 Starting Algorithm

An algorithm quoting the listed options has been provided in the file *options_mm_template.py* in the 'options_market_making' directory. However, it is incomplete. Your first challenge is to finish it. The reference implementation runs in a loop on a fixed interval, and takes the following actions:

- For each of the quoted options, update bid/ask orders in the options:
 1. Delete all existing outstanding orders
 2. Calculate the fair Black-Scholes option price, with the stock price input based on the current bid and ask of the ASML stock order-book and fixed volatility, interest rate, etc.
 3. Calculate the desired bid (buy) and ask (sell) prices by subtracting/adding a fixed offset (the bid-ask credit) to this price, and round away from the midpoint to the nearest tick. E.g., if the desired bid is 0.55, it is rounded to 0.50 and if the desired offer is 0.92 it is rounded up to 1.00. By adding this credit and rounding away from the mid we attempt to ensure the market making strategy captures some money to cover risks and returns a positive PnL.
 4. Insert limit orders on those levels for a specified volume, possibly adjusted to take into account the exchange position limits.
- Call the function to do a delta-hedge, which currently has no effect yet, as it remains for you to implement
- Wait some time until the next iteration

2.2 Finishing the Algorithm

Each of the following challenges requires to make a modification to the *template* file. Before making any changes, it is strongly recommended to make a copy of the original file to keep as reference. After that, run the starter implementation as it is. It is already running a strategy, can you follow along with what it does, how it works and spot the corresponding activity on the exchange visualizer?

- Make the bid-ask-credit asked for each option dynamic

Currently, the algorithm adds a fixed credit of 15 cents to each side of the theoretical option price to come to a desired bid and ask prices. This is suboptimal, the listed options are each different instruments with different characteristics due to their differing strikes and expiries, and hence they should be quoted with different bid-ask credit.

Modify the credit-logic to calculate a dynamic credit of your choice based on either:

- The current market conditions of the underlying stock,
- The current market conditions of the option,
- The greeks of the option, or
- A combination of the above.

Think about optimizing your PnL and Sharpe ratio, argue from first principles why yours is a good approach.



- Calculate the outstanding delta position

In the next sub-challenge you will implement a delta hedging strategy. For this, you will first need to calculate what your delta position actually is at each iteration. The algorithm already loops through the option positions and prints them. This can be extended to calculate the delta of the portfolio. Do so and confirm the delta positions that you see, per instrument and in aggregate, are sensible. Do you fully understand how to interpret these deltas? How do you take into account the position in the stock itself for your total portfolio delta position?

- Implement a delta hedge

As the started implementation does not delta hedge, it can run large risks to directional moves in the underlying stock. Explain how this risk could lead to losses. What is a worst-case scenario?

Implement the delta hedging functionality by hedging outstanding delta exposure using IOC orders in the stock. What is the benefit of using IOC orders for this hedge?

When finished, take a look at the exchange dashboard, do you see your delta position being hedged back to a value close to zero on each trade loop iteration?

- Increase quoting volumes

Currently, the algorithm only inserts orders for 3 lots, however the position limit per instrument is 100. So, it is possible to raise those volumes to increase the PnL.

If you raise the volumes too much however, that could lead to getting into positions of which the total delta is greater than 100 long/short. What is the potential issue with such an options position?

Either find a way to avoid this situation altogether or think of a clever way to deal with this risk. What approaches can you think of and what would be their relative Risk and PnL impacts? What do you consider the best one? Implement this.

2.3 (OPTIONAL) Free-Form Improvement

This exercise is optional, and requires some creative thinking. The algorithm is now functioning as intended and should be profitable in usual circumstances. But, given the dynamic nature of markets, this is just the beginning, further improvements are certainly still possible. Which direction will you choose to explore? Think about how your current strategy works and what appears to be a weak point or suboptimal in its competitiveness.

For the final part of this trading challenge, make an improvement of your choice which increases the algorithm's expected Sharpe ratio or PnL. If needed, feel free to make use of any instrument or exchange functionality not yet covered in the exercises. Reason about how your modification would function under different possible scenarios.